

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 3.4 MAGNETISM AND ELECTROMAGNETIC INDUCTION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the whole sub-strand. Answer the driving question: How does moving a wire through a magnetic field produce electricity, and how is this principle powering homes, farms and industries across Kenya? Use what you observed in our experiments, not just what you remember. Write in full sentences under each prompt. Use the words you learned, such as magnetic field lines, relative motion, induced e.m.f., galvanometer, coil turns and magnetic flux. Refer to the Kericho student, her bicycle dynamo and the school generator in your answers. Where a prompt asks for a diagram, draw and label it clearly. You may use your cumulative model and the Driving Question Board in your notes.

Part 1: Explaining the Kericho Puzzle

A Form One student in a boarding school in Kericho spins the wheel of a bicycle dynamo and a small LED lights up. When she holds the wheel still, the light goes out at once. There is no battery. (a) Explain why the LED lights up only when the wheel is spinning. (b) Explain what is happening to the magnetic field lines and the conductor while the light is on. (c) Use your galvanometer results from the electromagnetic induction experiment as evidence in your explanation.

Write your answer here:

Part 2: Why Faster Spinning Gives a Brighter Light

The Kericho student notices that when she spins the dynamo wheel faster, the LED becomes brighter. (a) Name the four factors that control the size of the induced e.m.f. and state, from your Lesson 5 observations, how each one changes the galvanometer deflection. (b) Using one of these factors, explain why faster spinning makes the LED brighter. (c) Explain how you made your Lesson 5 investigation a fair test.

Write your answer here:

Part 3: Magnets, Magnetic Fields and Why Stillness Matters

(a) Describe what you learned about magnetic field lines around a U-shaped magnet, including their direction, where they are crowded, and what the field is like between the poles. Explain why this region is important in a generator. **(b)** Describe two ways you saw a magnet lose its magnetism in Lesson 2 and explain, using magnetic domains, why this happens. **(c)** Explain why engineers at Kenya Power protect generator magnets from heat and vibration.

Write your answer here:

Part 4: From a Bicycle Dynamo to Kenya's National Grid

(a) Explain why the same principle that lights a bicycle dynamo LED can also power the dormitory in Kericho and Kenya's national grid. **(b)** Choose two applications of electromagnetic induction from our research (for example AC generators, transformers, Olkaria Geothermal Plant, Lake Turkana Wind Farm, electric bells, bicycle dynamos). For each, say how it uses induction and how it connects to the factors you studied. **(c)** Describe how the electric bell you built in Lesson 7 works, using the words magnetisation and demagnetisation.

Write your answer here:

Part 5: Final Answer to the Driving Question

Write a final answer, in a connected paragraph of about 8 to 10 sentences, to the driving question: How does moving a wire through a magnetic field produce electricity, and how is this principle powering homes, farms and industries across Kenya? Include: the name of the principle, what must happen for an e.m.f. to be induced, what happens when movement stops, the factors that make the e.m.f. larger, and at least two Kenyan examples of how this helps daily life (this answers the KICD question: How is the study of electromagnetic induction important in our daily life?). You may also add a labelled diagram of a generator or dynamo.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining electromagnetic induction	Clearly explains that relative motion causes the conductor to cut field lines, changing flux and inducing an e.m.f. that drives current. Explains why stillness gives zero e.m.f. and says no battery is needed.	States that movement between the conductor and the magnet induces an e.m.f. and that stopping stops the current, with limited explanation of field lines or flux.	Says only that spinning makes electricity or that the magnet makes it, without linking to relative motion or induced e.m.f.
Use of evidence from investigations	Uses specific results from the galvanometer experiments (stationary, parallel, upward, angled, speed, magnets, turns, 90° versus 0°) to support each claim in a Claim–Evidence–Reasoning way.	Uses some experimental results to support claims, but links to the reasoning may be incomplete.	Gives few or no observations from the experiments, or states results incorrectly.
Factors affecting induced e.m.f.	Names all four factors, correctly states how each changes the e.m.f., links them to Faraday's law and rate of change of flux, and explains fair testing.	Names at least three factors with mostly correct effects, with a limited link to Faraday's law.	Names one or two factors, or gives incorrect effects.
Magnetism concepts and applications in Kenya	Accurately describes field lines and domains, the bell's magnetisation and demagnetisation, and at least two	Describes most magnetism ideas and one or two applications, with small errors or missing detail.	Gives vague or incorrect descriptions and few or no Kenyan applications.

	Kenyan applications, including the role of transformers, using correct terms.		
Communication and connection to the driving question	Writes a coherent, well-organised final answer that links the Kericho phenomenon to the national grid and daily life, using scientific vocabulary accurately.	Writes a mostly clear answer that addresses the driving question, with some vocabulary slips or missing links.	Writes a disorganised or incomplete answer that does not fully address the driving question.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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